Controller Rebuild

Standalone reconstruction manual

Bare metal to serving DHCP, assuming nothing else works

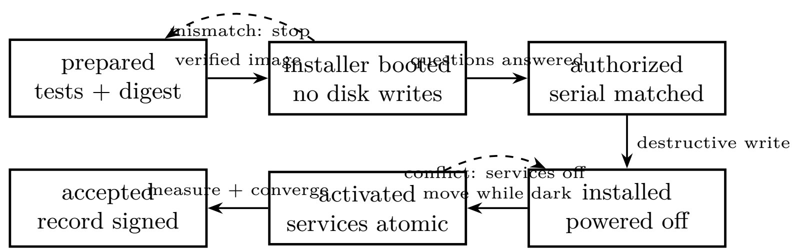
This manual assumes the Controller is dead and that nothing it hosted is available: no DNS, no PXE, no artifact service, no Git remote you can reach through it. It assumes you have this sheet on paper, an existing workstation, and the recorded inputs for the machine you are rebuilding. If any step here requires something the Controller was providing, that step is a defect — report it rather than working around it.

BEFORE YOU START, CONFIRM YOU HAVE ALL FIVE

1. This document, on paper or on a device that is not the Controller. **2.** A working Arch workstation to act as the bootstrap host. **3.** The installer image and its published SHA-256, or the ability to build it from this repository. **4.** The recorded instance values for this Controller: hostname, managed subnet, Controller address, DHCP pool, and the managed interface's MAC address. **5.** The LUKS2 recovery material, held off-host. Without it an existing encrypted disk is unrecoverable, and that is by design.

The rebuild state machine

Every arrow below has an observable boundary. Do not advance because elapsed time or familiar-looking output makes the next state seem likely.



Only recorded evidence moves the rebuild forward.

Stage 0 — Prepare the bootstrap host

GOVERNED BY ADR 0052

The bootstrap host runs the same dnsmasq and nginx configuration the Controller profile uses, generated by the same code, so the path you are about to exercise is the path you have already tested.

```
git clone <this repository> telos && cd telos/homelab
python3 -m unittest discover -s tests -t .
```

Expect: `OK`. If the test suite fails, stop. The installer's validation logic is what stands between an operator error and a wiped disk, and it is not optional.

Build or fetch the installer image and verify it:

```
sha256sum -c artifacts/manifest.sha256
```

Expect: Every line reports `OK`. A mismatch means the artifact is not the one that was tested — do not proceed. GOVERNED BY ADR 0043, ADR 0048

If not: Quarantine the mismatched image. Rebuild or reacquire it, then record both the expected and observed SHA-256; never repair the manifest to match an unexplained artifact.

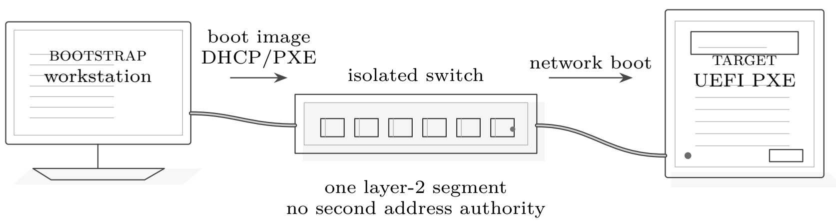
Answer before continuing: What exact image filename and SHA-256 will be entered in the acceptance record? Which interface is connected to the bootstrap segment, and which other interfaces are physically or administratively down?

`ip -br link`	Only the intended bootstrap interface is up on the bench segment; record its permanent MAC.
`sha256sum artifacts/<image>`	The 64 hexadecimal digits exactly match the separately obtained manifest value.

IF THE NETWORK ALREADY HAS A DHCP SERVER

Run the bootstrap dnsmasq in **ProxyDHCP** mode so it offers only boot options and never addresses. Two address-assigning DHCP authorities on one layer-2 network is forbidden at every stage, including this one. GOVERNED BY ADR 0008

Answer before continuing: Is the segment isolated, or is an existing DHCP authority present? Name the observation that proves the answer. If it is uncertain, use ProxyDHCP or stop; uncertainty never authorizes address service.



Bootstrap bench: verify the segment before the target requests a lease.

Stage 1 — Network boot the target

Set the target's firmware to UEFI network boot. The installer refuses a target booted in legacy BIOS or CSM mode and stops before offering any destructive action. GOVERNED BY ADR 0019

Expect: The Archiso environment reaches a shell and the installer starts. It has made no change to any disk at this point and will not until Stage 3.

If not: If PXE fails, keep the target in firmware or the read-only live environment. Check link, lease/ProxyDHCP traffic, TFTP handoff, HTTP status and image digest in that order. Do not change target disks while diagnosing boot.

`test -d /sys/firmware/efi`	Exit status zero proves UEFI boot; any failure means reboot and disable legacy/CSM.
`lsblk -o NAME,SIZE,MODEL,SERIAL,TYPE`	Inventory only: record the candidate disk tuple before the installer asks for authorization.

Stage 2 — Answer, and read the summary

GOVERNED BY ADR 0004, ADR 0045, ADR 0050

The installer collects, in order: the profile, the hostname, the target disk, the managed interface, and — when Controller network services are enabled — the four network inputs. It validates each as you enter it.

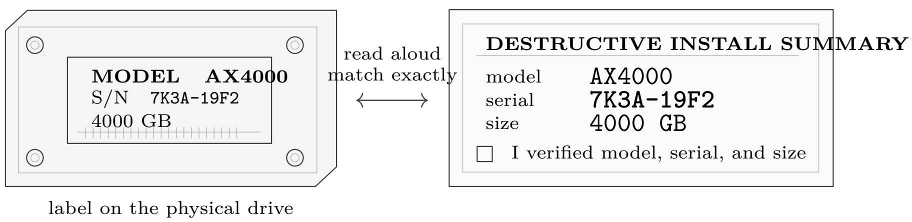
Input	What is checked
Target disk	Confirmed by model, serial and size. Never by `/dev/sdX`, because that is not stable across boots. Disks under 64 GiB are refused. GOVERNED BY ADR 0051
Managed interface	Chosen from an enumerated list showing kernel name, permanent MAC, link state and speed. There is no default. The MAC becomes the identity and is pinned to `lan0`.
`managed_ipv4_cidr`	Parses as IPv4, and names the network address itself — a CIDR with host bits set is rejected rather than silently normalised.
`controller_ipv4_address`	Usable unicast address inside the subnet, and *outside* the DHCP pool.
`dhcp_pool_start/end`	Both usable and inside the subnet; start not greater than end.

Everything else is derived and never prompted for: netmask, network and broadcast addresses, the client DNS server (the Controller), the suffix (`home.arpa`), and the absence of a default-router option.

THE PREFLIGHT SUMMARY IS THE AUTHORIZATION BOUNDARY
The summary states the exact disk that will be destroyed, by model and serial; the profile and hostname; and the complete entered and derived network plan. Read the disk serial aloud against the label on the drive. Physical access plus this confirmation *is* the authorization model — there is no second approval anywhere. Selecting a profile has authorized nothing.

Answer before continuing: Does the physical drive label match the summary's model, serial and size character for character? Is the managed-interface permanent MAC the one in the instance record? Is the Controller address outside the printed pool? If any answer is not an unqualified yes, which input will be corrected?

`lsblk -dn -o MODEL,SERIAL,SIZE <device>`	The live tuple matches both the physical label and the installer summary; save the output with the record.
`ip -br link`	The selected permanent MAC belongs to the connected managed interface and no interface was accepted by positional name alone.



Authorization is a physical comparison, never a device-name guess.

Stage 3 — Install

After confirmation the installer partitions, encrypts, creates filesystems, installs the base system with both kernels, writes systemd-boot and the UKIs, configures `lan0` and the static address, installs `sshd` with your public key, writes the manifest, syncs, and **powers off**. It does not reboot. GOVERNED BY ADR 0010

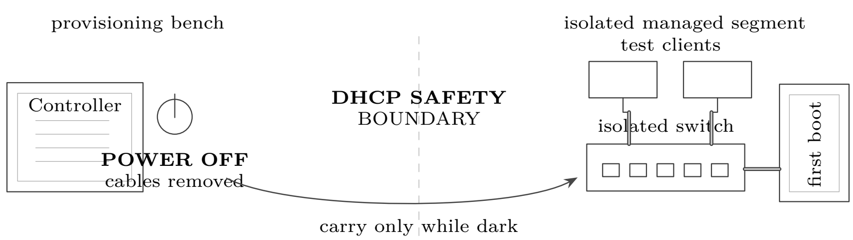
Expect: A clean poweroff, not a reboot. If the machine reboots into the installed system while still on the provisioning network, power it off before it can start DHCP.

If not: If installation stops before poweroff, photograph or capture the last successful phase and error, leave the target off, and inspect the log from the live environment. Never infer that an incomplete disk is bootable.

Answer before continuing: Has the installer displayed its final sync completion and requested no further input? Is the chassis visibly off before any cable is moved?

Stage 4 — Relocate, then first boot

Move the powered-off Controller to the isolated switch and power it on. Being powered off during the move is the primary DHCP-conflict safety boundary; it is physical, not a configuration flag.



Relocation sequence: off, disconnect, move, connect, then boot.

Under `development-proof` mode you will be asked for the LUKS2 passphrase at every boot. That is expected — Milestone A enrolls no TPM token. GOVERNED BY ADR 0043

First-boot activation verifies, before starting anything: that `lan0` exists, carries the expected MAC and has carrier; that the static address is configured; and that no other DHCP authority answers on the segment.

Expect: `systemctl is-active dnsmasq nginx` reports `active` for both. If activation failed, both are stopped and `journalctl -u homelab-first-boot` states which check failed. A partial start is a failure. GOVERNED BY ADR 0009, ADR 0012

If not: Keep both services stopped. Capture `journalctl -u homelab-first-boot -b` and the competing DHCP server's MAC, if reported. Remove the conflict or correct the interface/address record; do not start either daemon by hand.

Answer before continuing: Is the Controller now cabled only to the intended isolated segment? What observation shows that no other DHCP authority answered before activation? Are both services stopped if any preflight check failed?

Stage 5 — Prove it

From a separate client on the isolated switch:

```
ip addr show          # a lease inside the configured pool
ip route show         # no default route
dig +short ⟨hostname⟩.home.arpa @⟨controller-address⟩
```

`ip -4 -o addr show scope global`	Exactly one managed address, inside the configured pool; record address, prefix and client MAC.
`ip -4 route show default`	No output and exit status zero. Any default route is an acceptance failure.
`dig +short <host>.home.arpa @<controller>`	Exactly the recorded Controller address, with no extra answer.
`sha256sum /tmp/ipxe`	After fetching from the printed boot URL, HTTP succeeded and the observed digest matches the artifact manifest.

Lease	Inside the pool, and never the Controller's own address.
No default route	The client must have none. A default route here means dnsmasq advertised a router that does not exist, and every client on this segment is black-holed. GOVERNED BY ADR 0011
DNS	Returns the Controller's address.
PXE	A second target network boots from this Controller.
Idempotence	A second Ansible converge run reports no changes.

Stage 6 — Converge

GOVERNED BY ADR 0053

Everything beyond the base install is Ansible over SSH from this repository. Run it in check mode first and read the diff.

```
ansible-playbook -i ⟨inventory⟩ controller.yml -check -diff
ansible-playbook -i ⟨inventory⟩ controller.yml
ansible-playbook -i ⟨inventory⟩ controller.yml -check
```

Expect: The third run reports zero changed tasks. A role that is not idempotent is a role that will surprise you during an outage.

If not: Do not repeat the mutating run until it happens to look clean. Save the first and check-mode outputs, identify the task that remains changed or failed, correct its declared state, then restart with check mode.

Answer before continuing: Does check mode propose only intended changes? Can each destructive or service-restarting task be tied to the reviewed commit? Is the break-glass public key present while its private half remains off the Controller?

Acceptance record

Do not call the Controller rebuilt until this record is complete. Attach command output rather than transcribing favorable fragments.

Evidence	Record	Pass condition
Build identity	commit; image filename; SHA-256	Commit reviewed; digest matches manifest.
Target identity	model; serial; size; interface MAC	Physical label, live inventory and authorization summary agree.
Network plan	subnet; address; pool; client lease	Address outside pool; lease inside; no default route.
Activation	first-boot journal; service states	Preflight passed and both services active, or neither active.
Artifacts	boot URL; observed SHA-256	Fetch succeeds; digest matches.
Convergence	first-run recap; second check recap	Intended changes applied; final changed count is zero.
Recovery	recovery UKI date; custody check	Non-destructive boot proved; off-host material located without exposing it.

Operator: ______________________ Date/time: ______________________

Witness/reviewer: ______________________ Record ID: ______________________

Attachments retained with this record

Attachment	Filename or record reference	Observed digest / result
Installer manifest and image		
Authorization summary and disk inventory		
First-boot journal and client proof		
Apply recap and final check recap		

Recovering instead of rebuilding

Bad update Boot the recovery UKI, restore the operating-system checkpoint and its matching ESP artifact set, reboot. The checkpoint is taken before the transaction, not after. GOVERNED BY ADR 0028

Boot broken The recovery UKI is self-contained and boots without the installed root. It starts non-destructive and requires an explicit action before mounting anything writable. GOVERNED BY ADR 0024

TPM unlock fails Milestone B only. Early boot falls back to the recovery key, which is held off-host. Test that path *before* enabling TPM unlock, never after. GOVERNED BY ADR 0029

Disk dead Rebuild from this manual. Application data restores from its own backup, on its own schedule — it is not in the operating-system checkpoint. GOVERNED BY ADR 0027, ADR 0054

Recovery key lost The data is gone. This is the intended behaviour of encryption at rest, and it is why custody is a decision with its own ADRs rather than an afterthought.

Test this manual by using it

The QEMU acceptance matrix runs this entire sequence unattended against virtual hardware and asserts every expectation above. Any change to the installer, the profile or these instructions must pass it before touching real metal. A reconstruction manual nobody has followed end to end is a hypothesis. GOVERNED BY ADR 0056